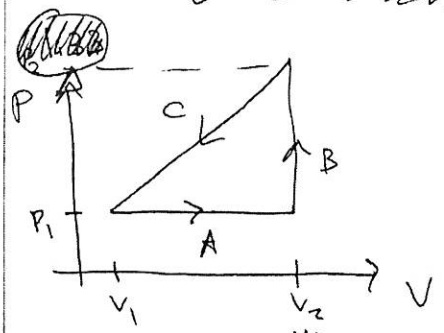
P 1.33



$PV = N k_B T$

(A) $W = +\int_{V_1}^{V_2} P\,dV = +P_1(V_2 - V_1) < 0$ work done on the gas

$U = N \cdot \frac{3}{2} k_B T \Rightarrow \Delta U = U_2 - U_1 = \frac{P_1}{N k_B}(V_2 - V_1) \cdot \frac{3}{2} k_B N$

$T_1 = \frac{P_1 V_1}{N k_B}, \; T_2 = \frac{P_2 V_2}{N k_B}$

$\Delta U = W + Q \Rightarrow \frac{3}{2} P_1 (V_2 - V_1) = -P_1(V_2 - V_1) + Q$

$Q_A = \frac{5}{2} P_1 (V_2 - V_1)$

(B) $\Delta V = 0 \Rightarrow W_B = 0 \quad T_i = \frac{P_1 V_2}{N k_B} \quad T_f = \frac{P_2 V_2}{N k_B}$

$\Delta U = \frac{3}{2} k_B N (T_f - T_i) = \frac{3}{2} V_2 (P_2 - P_1) > 0$

$\Rightarrow Q_B = \Delta U = \frac{3}{2} V_2 (P_2 - P_1)$

(C) $P = P_1 + \frac{V - V_1}{V_2 - V_1}(P_2 - P_1)$

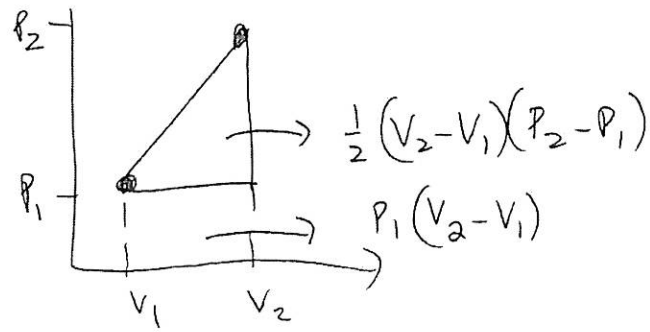
$\left.\begin{array}{ll} P_i = P_2 & V_i = V_2 \\ P_f = P_1 & V_f = V_1 \end{array}\right\} \quad W = +\int_{V_2}^{V_1} P\,dV = +\int_{V_2}^{V_1} \left(P_1 + \frac{P_2 - P_1}{V_2 - V_1}(V - V_1)\right) dV$

$= +P_1(V_1 - V_2) + \frac{P_2 - P_1}{V_2 - V_1}\left(\frac{1}{2} V^2 \Big|_{V_2}^{V_1} - V_1(V_1 - V_2)\right) =$

$= +P_1(V_1 - V_2) + \frac{P_2 - P_1}{V_2 - V_1}\underbrace{\left(\frac{1}{2} V_1^2 - \frac{1}{2} V_2^2 - V_1^2 + V_1 V_2\right)}_{-\frac{1}{2}(V_1^2 + V_2^2 - 2V_1V_2) = -\frac{1}{2}(V_1 - V_2)^2}$

$$W = +P_1(V_2 - V_1) + \frac{P_2 - P_1}{V_2 - V_1}\left(-\frac{1}{2}\right)(V_2 - V_1)^2 =$$

$$= P_1(V_2 - V_1) + \frac{1}{2}(P_2 - P_1)(V_2 - V_1) = \frac{1}{2}(P_2 + P_1)(V_2 - V_1)$$



$$U_f = \frac{3}{2} N k_B T_f = \frac{3}{2} N k_B \frac{P_1 V_1}{N k_B} = \frac{3}{2} P_1 V_1$$

$$U_i = \frac{3}{2} N k_B T_i = \frac{3}{2} N k_B \frac{P_2 V_2}{N k_B} = \frac{3}{2} P_2 V_2$$

$$\Delta U = U_f - U_i = \frac{3}{2}(P_1 V_1 - P_2 V_2)$$

$$\Rightarrow Q_C = -\frac{3}{2}(P_2 V_2 - P_1 V_1) - \frac{1}{2}(P_2 + P_1)(V_2 - V_1) =$$

$$= -\frac{3}{2}(P_2 V_2 - P_1 V_1) - \frac{1}{2}(P_2 V_2 - P_2 V_1 + P_1 V_2 - P_1 V_1)$$

$$= -2P_2 V_2 + 2P_1 V_1 + \frac{1}{2}(P_2 V_1 - P_1 V_2)$$

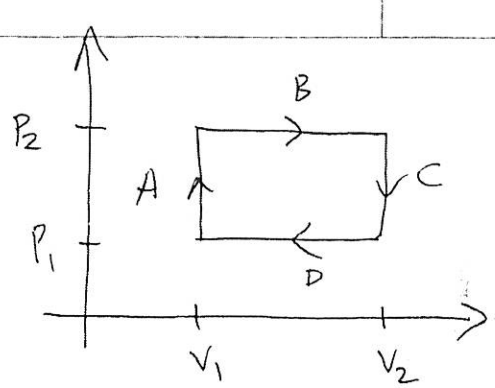
Total work: $\sum W = W_A + \underset{\overset{\shortparallel}{0}}{W_B} + W_C = -P_1(V_2 - V_1) + \frac{1}{2}(P_2 + P_1)(V_2 - V_1) =$

$$= \frac{1}{2}(P_2 - P_1)(V_2 - V_1)$$

Internal energy did not change as initial and final P, V are the same

$\Rightarrow \sum W + \sum Q = 0 \Rightarrow$ system does work and emits heat.

P 1.34



diatomic gas $\Rightarrow$ add rotational degrees of freedom

$$E = \underset{\frac{3}{2}kT}{\underset{\parallel}{K.E.}} + \underset{2\cdot\left(\frac{1}{2}kT\right)}{\underset{\parallel}{K.ROT}} = \frac{5}{2}kT.$$

$$\text{Total } \Delta W = \oint P\,dV = (P_2 - P_1)\,\overset{=0}{\Delta V_A} + P_2(V_2 - V_1) + (P_1 - P_2)\,\underset{0}{\underset{\parallel}{\Delta V_C}}$$

$$+ P_1(V_1 - V_2) = (P_2 - P_1)(V_2 - V_1).$$

Total $\Delta U = 0$ because initial and final states are the same

Total $\Delta Q = (P_2 - P_1)(V_2 - V_1)$ because $\Delta U = \Delta Q - \Delta W = 0$.

$U = \frac{5}{2} kT \cdot N$, $PV = NkT \Rightarrow U = \frac{5}{2} PV$

$$\Rightarrow \left.\begin{aligned} U_B - U_A &= \tfrac{5}{2}(P_2 - P_1)V_1 \\ U_C - U_B &= \tfrac{5}{2} P_2 (V_2 - V_1) \\ U_D - U_C &= \tfrac{5}{2}(P_1 - P_2)V_2 \\ U_A - U_D &= \tfrac{5}{2} P_1 (V_1 - V_2) \end{aligned}\right\} \Delta U = 0$$

(P3) Distribution of velocities in two dimensions should be

$$f(\vec{v})\,dv_x\,dv_y = C_2\, e^{-\frac{m}{2kT}(v_x^2+v_y^2)}\,dv_x\,dv_y$$

Normalization C_2 is determined by

$$\int f(\vec{v})\,dv_x\,dv_y = 1 \Rightarrow \text{Us}\; dv_x\,dv_y = 2\pi v\,dv\,,\quad v=\sqrt{v_x^2+v_y^2}$$

$$2\pi C_2 \int_0^\infty dv\, v\, e^{-\frac{mv^2}{2kT}} = \pi C_2 \int_0^\infty d(v^2)\, e^{-\frac{mv^2}{2kT}} =$$

$$=\left(x=\frac{mv^2}{2kT}\right) = \pi C_2 \frac{2kT}{m}\int_0^\infty dx\, e^{-x} = \frac{2\pi kT}{m} C_2 = 1$$

$$\Rightarrow C_2 = \frac{m}{2\pi kT}$$

Average speed $\langle v\rangle = \dfrac{\int v\, f(\vec{v})\,dv_x\,dv_y}{\int f(\vec{v})\,dv_x\,dv_y} =$

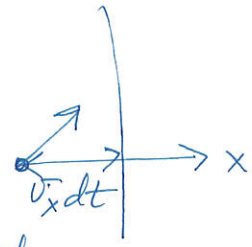
$$= \frac{m}{2\pi kT}\int_0^\infty dv\cdot v\cdot 2\pi\cdot v\, e^{-\frac{mv^2}{2kT}} =$$

$$= \frac{m}{kT}\int_0^\infty dv\cdot v^2 e^{-\frac{mv^2}{2kT}} = \left(\text{let } v=\sqrt{\frac{2kT}{m}}\,x\right) =$$

$$= \frac{m}{kT}\left(\frac{2kT}{m}\right)^{3/2}\underbrace{\int_0^\infty dx\, x^2 e^{-x^2}}_{\frac{\sqrt{\pi}}{4}} = \frac{\sqrt{\pi}}{2}\sqrt{\frac{2kT}{m}} = \sqrt{\frac{\pi k_B T}{2m}}\,.$$

P4 Number of collisions with area A, per unit time?

$$(v_x dt) f(\vec{p})\, dp_x\, dp_y\, dp_z$$



Particle, with $\vec{p} = (p_x, p_y, p_z)$, which is at distance $v_x dt$ from the wall, will collide with that wall after time dt.

$\Rightarrow$ total number of particles colliding with wall per unit time

$$d\nu = \underbrace{\left(\frac{N}{V}\right)}_{=n} \underbrace{v_x}_{=\frac{p_x}{m}} \frac{e^{-p_x^2/2mkT} \int_{-\infty}^{\infty} dp_y\, dp_z\, e^{-\frac{p_y^2+p_z^2}{2mkT}}}{(2\pi mkT)^{3/2}}$$

$$\Rightarrow \nu = \frac{n}{m} \frac{\int_0^{\infty} dp_x\, p_x e^{-p_x^2/2mkT}}{(2\pi mkT)^{1/2}}$$

$$= \frac{n}{2m}(2mkT) \int_0^{\infty} \frac{du\, e^{-u}}{(2\pi mkT)^{1/2}} = \frac{n}{m}\sqrt{\frac{mkT}{2\pi}} =$$

$$= \frac{1}{4} n \sqrt{\frac{16kT}{2\pi m}} = \frac{1}{4} n \cdot \bar{v} = \nu \left(= \#\text{ particles hitting unit area per unit time}\right)$$

here $\bar{v} = \sqrt{\frac{8kT}{\pi m}} = \langle |\vec{v}| \rangle$

is the average speed of 3d Maxwell's gas at temperature T.